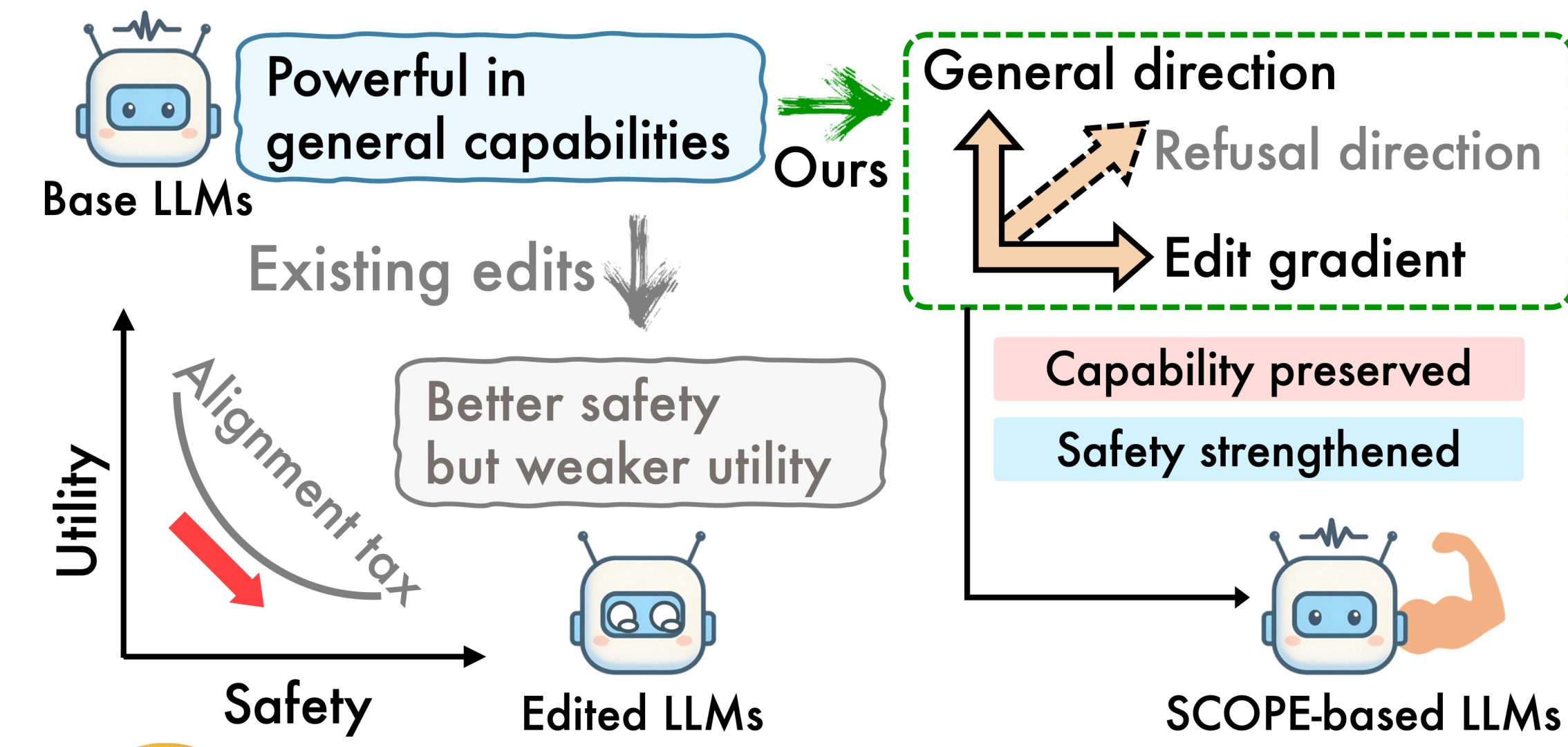


## Introduction



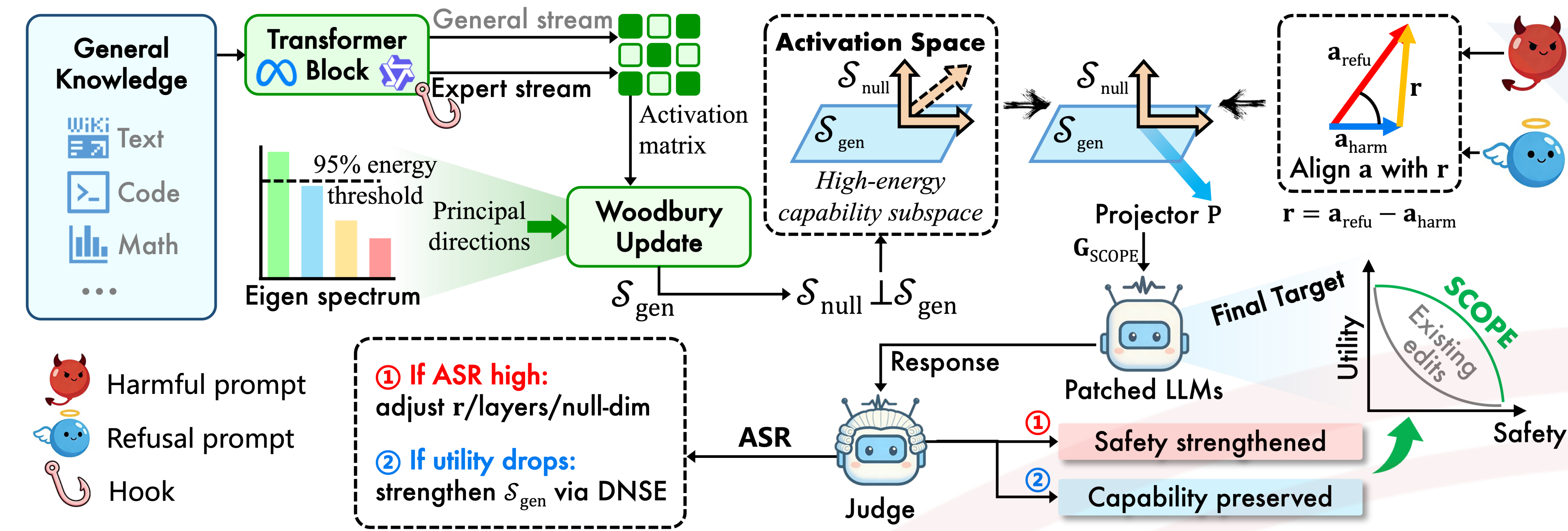
- ✗ Current post-hoc safety editing methods lack explicit invariance targets, causing **cumulative utility drift under sequential patches**.
- ✗ Existing null-space constrained approaches enforce parameter locality but **fail to guarantee functional preservation of general capabilities**.
- ✗ Conventional alignment methods entangle safety objectives with general-purpose representations, **imposing a significant alignment tax**.

## Motivation

- ✓ Deployed LLMs require **continual safety updates** that strengthen refusal behaviors without degrading general capabilities.
- ✓ Post-hoc editing lacks a geometric framework that **decouples capability preservation** from safety-specific modifications.
- ✓ Real-world deployment demands **low-latency, retraining-free** safety patching to rapidly adapt to emerging threats.

## Experiments

## Method



### Discover: Capability subspace identification.

Given a general-purpose corpus  $\mathcal{D}_{\text{cap}}$ , collect layer-wise activation vectors  $\mathbf{h}_i^{(l)} \in \mathbb{R}^{d_l}$  and construct a regularized empirical covariance matrix:

$$\mathbf{C}^{(l)} = \lambda \mathbf{I}_{d_l} + \sum_{i=1}^{n_l} \mathbf{h}_i^{(l)} \mathbf{h}_i^{(l)\top}.$$

Eigendecomposition of  $\mathbf{C}^{(l)}$  yields eigenvectors sorted by descending eigenvalues. To balance capability preservation against editing flexibility, the truncation rank  $r$  is determined by an energy threshold while strictly bounded above by  $d_l - d_{\min}$  to maintain a minimum editable null-space dimension:

$$r = \min\{r_{\text{energy}}, d_l - d_{\min}\}.$$

The top- $r$  eigenvectors  $\mathbf{U}^{(l)} \in \mathbb{R}^{d_l \times r}$  span the **general capability subspace**  $\mathcal{S}_{\text{gen}}$ . This induces complementary projectors:

$$\mathbf{P}^{(l)} = \mathbf{U}^{(l)} \mathbf{U}^{(l)\top}, \quad \mathbf{N}^{(l)} = \mathbf{I}_{d_l} - \mathbf{P}^{(l)},$$

where  $\mathbf{P}^{(l)}$  projects onto  $\mathcal{S}_{\text{gen}}$  and  $\mathbf{N}^{(l)}$  projects onto its orthogonal complement, the **editable null space**  $\mathcal{S}_{\text{null}}$ .

### Govern: Null-space constrained refusal injection.

A contrastive anchor set  $\{(c_i^{\text{harm}}, c_i^{\text{refu}})\}_{i=1}^N$  captures the activation shift from harmful intent to refusal. The layer-wise refusal direction is:

$$\mathbf{r}^{(l)} = \frac{\frac{1}{N} \sum_{i=1}^N (a^{(l)}(c_i^{\text{refu}}) - a^{(l)}(c_i^{\text{harm}}))}{\left\| \frac{1}{N} \sum_{i=1}^N (a^{(l)}(c_i^{\text{refu}}) - a^{(l)}(c_i^{\text{harm}})) \right\|_2}.$$

The refusal direction is projected onto the null space via  $\mathbf{N}^{(l)}$  and then renormalized as  $\hat{\mathbf{r}}^{(l)} = \mathbf{N}^{(l)} \mathbf{r}^{(l)} / \|\mathbf{N}^{(l)} \mathbf{r}^{(l)}\|_2$ . Next, the optimal editing layer is selected by maximizing refusal signal density per editable dimension:  $s^{(l)} = \|\mathbf{N}^{(l)} \mathbf{r}^{(l)}\|_2^2 / \text{Tr}(\mathbf{N}^{(l)})$ . At the chosen layer, a raw gradient is computed from safety loss and KL regularization as:  $\mathbf{G}_{\text{raw}}^{(l)} = \nabla_{\mathbf{W}^{(l)}} (\mathcal{L}_{\text{safe}} + \beta \mathcal{L}_{\text{KL}})$ . This gradient is first projected onto the null space, then confined to the refusal direction as follows:

$$\mathbf{G}_{\text{null}}^{(l)} = \mathbf{G}_{\text{raw}}^{(l)} \mathbf{N}^{(l)}, \quad \mathbf{G}_{\text{SCOPE}}^{(l)} = \mathbf{G}_{\text{null}}^{(l)} (\hat{\mathbf{r}}^{(l)} \hat{\mathbf{r}}^{(l)\top}).$$

The weight update is  $\Delta \mathbf{W}^{(l)} \leftarrow \Delta \mathbf{W}^{(l)} \leftarrow \eta \mathbf{G}_{\text{SCOPE}}^{(l)}$ , ensuring edits remain strictly within  $\mathcal{S}_{\text{null}}$  and aligned with the refusal trajectory.

### Expand: Dynamic null-space expansion.

When a new domain batch  $\mathbf{X}^{(l)}$  arrives, compute its local second-moment matrix  $\mathbf{C}_{\text{dom}}^{(l)} = \lambda \mathbf{I} + \mathbf{X}^{(l)} \mathbf{X}^{(l)\top}$ . The energy coverage ratio quantifies alignment with the existing capability subspace as:  $\kappa = \text{Tr}(\mathbf{P}^{(l)} \mathbf{C}_{\text{dom}}^{(l)}) / \text{Tr}(\mathbf{C}_{\text{dom}}^{(l)})$ .

If  $\kappa < \tau_{\text{cov}}$ , where  $\tau_{\text{cov}}$  denotes a threshold for all layers, the subspace is updated by incorporating  $\mathbf{X}^{(l)}$  into the global covariance as:  $\mathbf{C}_+^{(l)} = \mathbf{C}^{(l)} + \mathbf{X}^{(l)} \mathbf{X}^{(l)\top}$ . Thus, the direct recomputation is avoided via the *Sherman-Morrison-Woodbury* identity:

$$(\mathbf{C}_+^{(l)})^{-1} = (\mathbf{C}^{(l)})^{-1} - (\mathbf{C}^{(l)})^{-1} \mathbf{X}^{(l)} \mathbf{M}^{(l)} \mathbf{X}^{(l)\top} (\mathbf{C}^{(l)})^{-1}, \quad \mathbf{M}^{(l)} = [\mathbf{I}_m + \mathbf{X}^{(l)\top} (\mathbf{C}^{(l)})^{-1} \mathbf{X}^{(l)}]^{-1}.$$

Eigendecomposition of  $(\mathbf{C}_+^{(l)})^{-1}$  extracts eigenvectors corresponding to its smallest eigenvalues, updating  $\mathbf{U}_+^{(l)}$ ,  $\mathbf{P}_+^{(l)}$ , and  $\mathbf{N}_+^{(l)}$  while preserving at least  $d_{\min}$  editable dimensions. This enables low-latency and continual expansion of the protected subspace under distribution shifts.

Table 1: Safety performance comparison via ASR.

| Model                      | Harmful Compliance Group |                  |                  |                  |                   | Red-Teaming Group |                  | Retraining-free |
|----------------------------|--------------------------|------------------|------------------|------------------|-------------------|-------------------|------------------|-----------------|
|                            | AdvB                     | JailbreakB       | PKU-S            | HarmB            | SORRY             | HarmQA            | ALERT            |                 |
| <b>Llama3-8B-Instruct</b>  | 1.36±0.30                | 2.00±0.08        | 3.29±0.17        | 7.50±0.41        | 34.15±3.59        | 1.89±0.39         | 3.81±0.22        | –               |
| + DPO-H [25, 48]           | 37.41±0.87               | 47.00±0.81       | 30.23±0.42       | 57.32±0.68       | 89.08±5.58        | 24.11±0.23        | 27.68±0.42       | ✗               |
| + DPO-S [25, 48]           | <b>0.06±0.09</b>         | 2.67±1.25        | 2.24±0.05        | 5.67±0.24        | 14.83±1.65        | 1.20±0.08         | 3.26±0.24        | ✗               |
| + DPO-Mix [48, 67]         | 3.84±0.42                | 6.67±1.25        | 3.96±0.40        | 15.58±0.42       | 23.26±2.31        | 3.40±0.02         | 5.21±0.12        | ✗               |
| + Safe-RLHF [13]           | 14.39±0.13               | 16.33±0.47       | 6.50±0.37        | 41.34±1.03       | 44.77±3.58        | 4.72±0.02         | 13.66±0.09       | ✗               |
| + PeCAN [25]               | 0.38±0.16                | 2.33±0.94        | 3.22±0.44        | 8.83±1.31        | <b>12.79±1.35</b> | 1.13±0.04         | 3.46±0.32        | ✓               |
| + MoCAN [25]               | 1.09±0.33                | 2.67±0.94        | 3.38±0.60        | 6.83±1.65        | 36.45±3.06        | 1.41±0.07         | 4.60±0.25        | ✓               |
| + W-DOOR [68]              | 0.75±0.12                | 4.33±0.94        | 3.31±0.22        | 2.81±0.18        | 30.46±3.40        | 0.66±0.11         | 1.86±0.06        | ✓               |
| + BFPO [67]                | 0.64±0.24                | 2.67±0.47        | <b>1.86±0.12</b> | 4.16±0.26        | 29.73±0.96        | <b>0.34±0.08</b>  | <b>7.22±0.06</b> | ✓               |
| + AlphaEdit [15]           | 0.55±0.10                | <b>2.22±0.35</b> | <b>1.85±0.12</b> | <b>2.10±0.18</b> | 16.80±2.10        | <b>0.82±0.10</b>  | <b>1.60±0.18</b> | ✓               |
| + <b>SCOPE (Ours)</b>      | <b>0.10±0.10</b>         | <b>1.07±0.63</b> | <b>1.92±0.16</b> | <b>1.06±0.04</b> | <b>11.96±1.94</b> | <b>0.10±0.06</b>  | <b>1.02±0.02</b> | ✓               |
| <b>Qwen2.5-7B-Instruct</b> | 1.28±0.09                | 3.00±0.81        | 2.95±0.06        | 13.83±1.43       | 41.88±0.85        | 3.26±0.30         | 8.62±0.43        | –               |
| + DPO-H [25, 48]           | 1.47±0.36                | 2.33±0.47        | 3.26±0.32        | 14.67±0.24       | 41.21±0.03        | 3.65±0.18         | 8.27±0.09        | ✗               |
| + DPO-S [25, 48]           | 1.67±0.24                | 2.67±0.94        | 3.11±0.05        | 18.33±0.47       | <b>40.91±0.32</b> | 2.83±0.23         | 9.61±0.27        | ✗               |
| + DPO-Mix [48, 67]         | 2.24±0.39                | 4.33±0.47        | 3.71±0.28        | 18.67±1.03       | 41.48±0.12        | 3.52±0.20         | 9.22±0.09        | ✗               |
| + Safe-RLHF [13]           | 6.64±0.48                | 25.00±0.81       | 3.86±0.12        | 32.25±0.53       | 37.95±0.10        | 2.76±0.08         | 15.24±0.02       | ✗               |
| + PeCAN [25]               | 1.99±0.48                | 5.67±0.47        | 3.55±0.14        | 18.48±0.02       | 45.42±0.03        | 3.42±0.10         | 8.65±0.37        | ✓               |
| + MoCAN [25]               | 1.22±0.18                | 2.00±0.00        | 3.51±0.40        | 17.83±2.09       | 41.33±0.02        | 3.55±0.23         | 8.64±0.14        | ✓               |
| + W-DOOR [68]              | <b>0.00±0.00</b>         | 3.33±1.25        | 3.72±0.03        | 0.51±0.01        | <b>15.00±0.03</b> | <b>0.31±0.01</b>  | 3.40±0.03        | ✓               |
| + BFPO [67]                | 0.96±0.31                | <b>1.33±0.94</b> | <b>1.35±0.05</b> | 11.33±0.85       | 34.31±0.01        | 0.74±0.13         | 5.83±0.30        | ✓               |
| + AlphaEdit [15]           | 0.40±0.10                | 2.20±0.40        | 1.90±0.10        | 1.20±0.28        | 19.80±1.10        | 0.90±0.12         | 1.30±0.18        | ✓               |
| + <b>SCOPE (Ours)</b>      | <b>0.30±0.10</b>         | <b>0.62±0.47</b> | <b>0.83±0.03</b> | <b>0.50±0.02</b> | <b>17.38±0.01</b> | <b>0.11±0.04</b>  | <b>0.74±0.20</b> | ✓               |

\* Results for all non-AlphaEdit methods are taken from Niu et al. [43]. Retraining-free denotes post-hoc editing without full backbone retraining.

Table 2: General capability retention comparison.

| Model               | IF         |            | STEM           |            | Math Reasoning & Adv. Solving |                |               | Coding       |               |
|---------------------|------------|------------|----------------|------------|-------------------------------|----------------|---------------|--------------|---------------|
|                     | WR (%) ↑   | AlpacaEval | Accuracy (%) ↑ |            | GSM8K                         | Accuracy (%) ↑ |               | Pass@1 (%) ↑ | LiveCodeBench |
|                     |            |            | MMLU           | SuperGPQA  |                               | MATH           | OlympiadBench |              |               |
| Llama3-8B-Instruct  | 96.15±0.64 | 63.79±0.38 | 23.18±0.03     | 75.44±1.26 | 27.55±0.05                    | 5.70±0.40      | 13.37±0.09    |              |               |
| + DPO-H [25, 48]    | 91.61±0.88 | 60.47±0.40 | 18.77±0.04     | 59.97±1.33 | 22.26±0.05                    | 3.85±0.45      | 10.48±0.11    |              |               |
| + DPO-S [25, 48]    | 93.60±0.77 | 57.78±0.40 | 21.11±0.06     | 62.55±1.33 | 22.85±0.05                    | 3.45±0.15      | 10.71±0.07    |              |               |
| + DPO-Mix [48, 67]  | 93.85±0.81 | 60.48±0.39 | 19.62±0.06     | 63.76±1.32 | 22.90±0.10                    | 3.80±0.20      | 11.24±0.10    |              |               |
| + Safe-RLHF [13]    | 52.55±1.75 | 58.91±0.39 | 15.50±0.01     | 51.71±1.38 | 13.20±0.00                    | 3.30±0.01      | 8.96±0.13     |              |               |
| + PeCAN [25]        | 88.45±1.39 | 60.44±0.39 | 20.15±0.06     | 59.36±1.35 | 21.19±0.01                    | 4.50±0.20      | 11.44±0.11    |              |               |
| + MoCAN [25]        | 95.71±0.76 | 63.88±0.38 | 23.19±0.04     | 74.98±1.19 | 26.80±0.10                    | 5.10±0.40      | 13.34±0.12    |              |               |
| + W-DOOR [68]       | 91.61±0.97 | 62.50±0.39 | 22.19±0.03     | 71.11±1.25 | 20.11±0.09                    | 5.25±0.13      | 12.68±0.08    |              |               |
| + BFPO [67]         | 97.20±0.37 | 63.98±0.38 | 23.18±0.09     | 78.32±1.14 | 28.10±0.20                    | 5.49±0.01      | 13.13±0.05    |              |               |
| + AlphaEdit [15]    | 96.30±0.84 | 63.60±0.32 | 22.88±0.05     | 76.45±1.10 | 27.95±0.14                    | 5.50±0.18      | 13.05±0.08    |              |               |
| + SCOPE (Ours)      | 96.75±0.93 | 63.76±0.20 | 22.73±0.03     | 77.22±1.33 | 29.20±0.30                    | 5.31±0.30      | 13.27±0.12    |              |               |
| Qwen2.5-7B-Instruct | 94.35±0.41 | 71.71±0.36 | 27.72±0.09     | 81.65±1.07 | 74.80±0.15                    | 37.80±0.24     | 24.07±0.08    |              |               |
| + DPO-H [25, 48]    | 94.41±0.80 | 71.66±0.36 | 26.62±0.02     | 80.21±1.10 | 74.05±0.05                    | 36.00±0.01     | 24.87±0.13    |              |               |
| + DPO-S [25, 48]    | 94.47±0.80 | 71.35±0.36 | 27.17±0.01     | 80.67±1.09 | 74.27±0.05                    | 35.63±0.05     | 26.09±0.05    |              |               |
| + DPO-Mix [48, 67]  | 93.60±0.86 | 71.56±0.36 | 27.06±0.02     | 81.35±1.07 | 74.40±0.10                    | 38.45±0.05     | 25.55±0.11    |              |               |
| + Safe-RLHF [13]    | 57.02±1.73 | 70.77±0.36 | 18.67±0.01     | 75.28±1.19 | 44.63±0.03                    | 22.22±0.24     | 21.84±0.09    |              |               |
| + PeCAN [25]        | 94.35±0.41 | 71.83±0.36 | 27.47±0.04     | 82.87±1.04 | 75.20±0.10                    | 35.95±0.05     | 23.57±0.13    |              |               |
| + MoCAN [25]        | 95.28±0.74 | 71.71±0.36 | 27.48±0.02     | 81.65±0.97 | 75.50±0.10                    | 39.65±0.03     | 24.16±0.04    |              |               |
| + W-DOOR [68]       | 75.96±1.30 | 70.25±0.37 | 18.58±0.01     | 57.85±1.36 | 37.15±0.14                    | 15.15±0.12     | 23.23±0.07    |              |               |
| + BFPO [67]         | 95.22±0.74 | 71.72±0.36 | 27.98±0.13     | 83.85±1.01 | 74.55±0.03                    | 36.35±0.06     | 24.36±0.04    |              |               |
| + AlphaEdit [15]    | 95.20±0.78 | 71.40±0.38 | 27.98±0.05     | 83.80±1.00 | 75.90±0.08                    | 39.60±0.10     | 25.60±0.07    |              |               |
| + SCOPE (Ours)      | 95.32±0.82 | 71.77±0.36 | 27.79±0.04     | 83.07±1.12 | 75.94±0.03                    | 39.25±0.04     | 26.03±0.04    |              |               |

\* Results for all non-AlphaEdit methods are taken from Niu et al. [43]. WR denotes the win rate on ALPACA-EVAL 2.0.

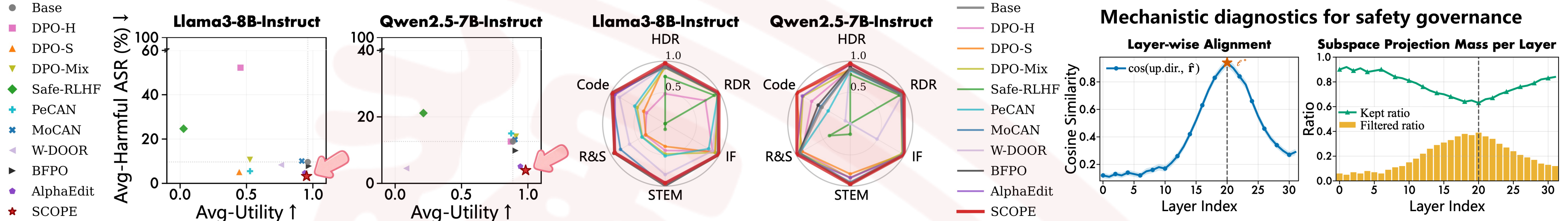


Table 3: 70B-scale generalization.

| Metric                                               | Llama3-70B-Instruct | +SCOPE            | Qwen2.5-72B-Instruct | +SCOPE            |
|------------------------------------------------------|---------------------|-------------------|----------------------|-------------------|
| <b>Safety Benchmarks - ASR (%) ↓</b>                 |                     |                   |                      |                   |
| AdvB                                                 | 0.72±0.14           | <b>0.18±0.08</b>  | 0.52±0.11            | <b>0.12±0.06</b>  |
| JailbreakB                                           | 1.36±0.22           | <b>0.18±0.12</b>  | 0.68±0.20            | <b>0.20±0.10</b>  |
| PKU-S                                                | 2.14±0.16           | <b>0.34±0.09</b>  | 1.58±0.12            | <b>0.46±0.08</b>  |
| HarmB                                                | 4.80±0.32           | <b>0.88±0.12</b>  | 5.56±1.02            | <b>0.92±0.22</b>  |
| SORRY                                                | 22.48±2.34          | <b>9.42±2.08</b>  | 26.66±0.76           | <b>10.80±0.12</b> |
| HarmQA                                               | 1.18±0.14           | <b>0.22±0.07</b>  | 1.36±0.15            | <b>0.26±0.08</b>  |
| ALERT                                                | 2.62±0.28           | <b>0.60±0.22</b>  | 3.42±0.43            | <b>0.78±0.34</b>  |
| <b>Utility Benchmarks - Accuracy/WR/Pass@1 (%) ↑</b> |                     |                   |                      |                   |
| AlpacaEval                                           | 97.80±0.81          | <b>97.90±0.82</b> | 97.40±0.82           | <b>97.60±0.80</b> |
| MMLU                                                 | 79.00±0.32          | 78.90±0.20        | <b>85.32±0.36</b>    | 84.92±0.28        |
| SuperGPQA                                            | 39.00±0.18          | 38.70±0.16        | <b>47.20±0.22</b>    | 46.90±0.20        |
| GSM8K                                                | 89.80±0.94          | <b>90.20±0.90</b> | 94.90±0.85           | <b>95.10±0.88</b> |
| MATH                                                 | 47.80±0.20          | <b>48.60±0.22</b> | 79.80±0.18           | <b>80.60±0.16</b> |
| OlympiadBench                                        | 23.80±0.05          | <b>24.20±0.42</b> | 50.20±0.30           | <b>50.60±0.32</b> |
| LiveCodeBench                                        | 27.00±0.12          | <b>27.10±0.08</b> | 33.52±0.20           | <b>33.74±0.04</b> |

Sequential domain expansion

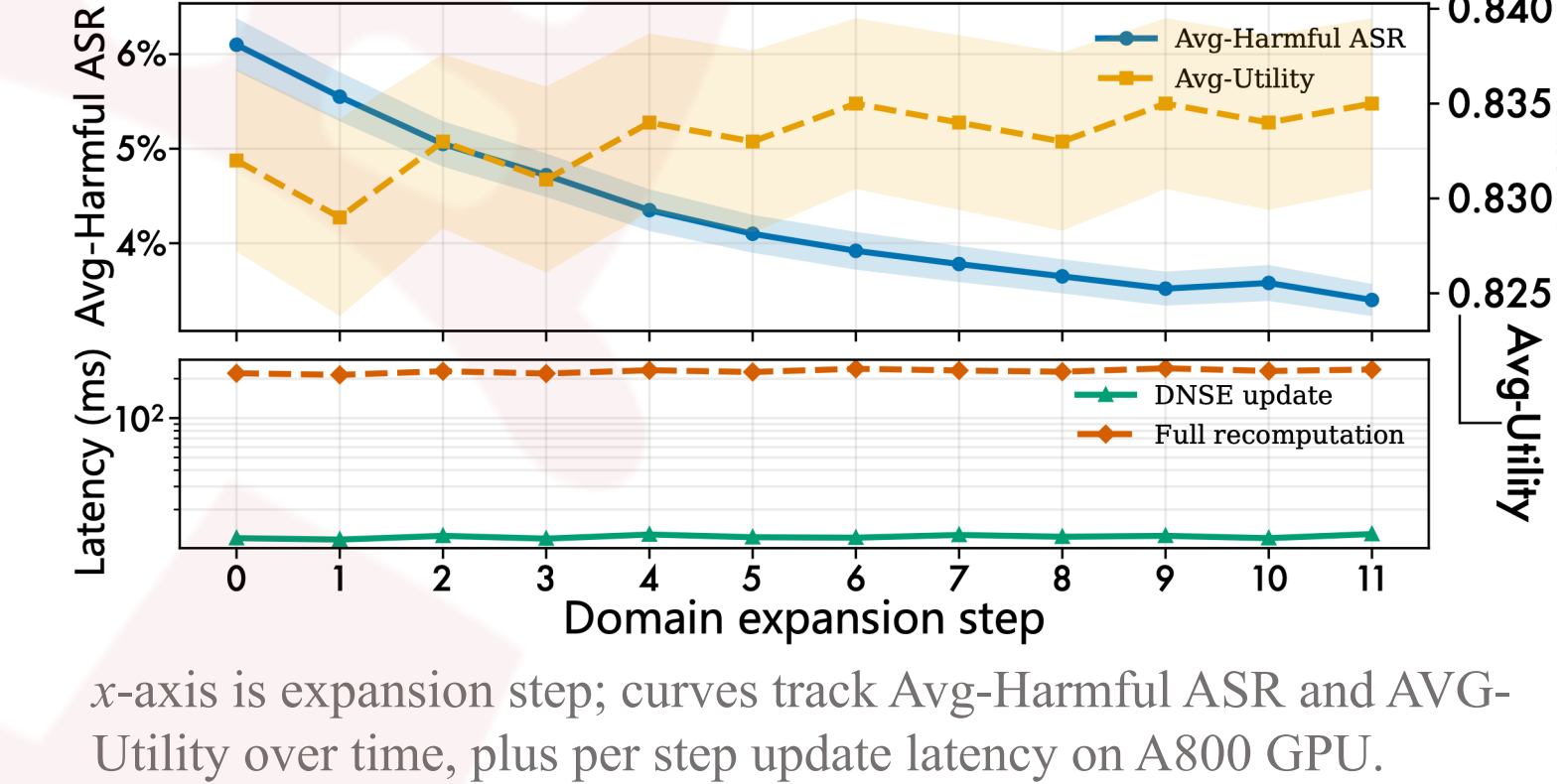


Table 4: Ablation studies on Qwen2.5-7B-Instruct.

(a) Component ablation: Effects of *Discover-Govern-Expand* pipeline in SCOPE, reporting Avg-Harmful ASR (A.H., %), Avg-Red-Team ASR (A.R., %), Avg-Utility (A.U.), and per step update latency (Lat., ms) on A800. (b) Online maintenance: Woodbury-style incremental updates vs. full recomputation under a fixed expansion schedule, reporting end-stage Avg-Harmful ASR (End A.H., %), Avg-Utility (End A.U.), cumulative maintenance time (C.Time, s), and peak maintenance memory (P.M., GB). C.Time and P.M. denote the additional overhead excluding the model’s resident footprint.

| (a) Component ablation |        |        |        |        | (b) Online maintenance |          |         |
|------------------------|--------|--------|--------|--------|------------------------|----------|---------|
| w/o                    | A.H. ↓ | A.R. ↓ | A.U. ↑ | Lat. ↓ | Metric                 | Woodbury | Recomp. |
| Dsc.                   | 3.12   | 0.44   | 0.620  | 10     | End A.H. ↓             | 3.55     | 3.53    |
| Gov.                   | 6.80   | 2.10   | 0.940  | 18     | End A.U. ↑             | 0.968    | 0.967   |
| Exp.                   | 4.45   | 0.82   | 0.952  | 220    | C.Time ↓               | 0.09     | 1.8     |